

AI Ops assignment 1

K Kevin Sona (DA24B007)

August 30, 2026

Question 1

1. Technical debt categories:

- a) Changing anything changes everything (CACE)
- b) Undeclared consumers
- c) Configuration & Glue-Code Debt

2. The glue code debt in case (c) can be mitigated with MLFlow, which allows us to track all training configurations such as hyperparameters in one place.

Question 2

Ran a total of 12 experiments with 3 learning rate, 2 width and 2 depth configurations.

The best-performing runs were mlp-lr0.001-w128-d2 and mlp-lr0.001-w128-d1, both reaching 0.977 accuracy and 0.977 macro-F1. The two-layer model converged faster (16 vs 19 epochs) at the same final training loss of 0.012, making it marginally better. Both combine a moderate learning rate with the widest hidden layer, converging well before the 200 epoch ceiling.

Learning rate had the largest effect. Mean accuracy was 0.975 at 1e-3, 0.971 at 1e-2, and 0.971 at 1e-4, giving a spread of 0.4 points. Width contributed roughly 0.3 points and depth under 0.1.

The train loss and validation accuracy trends do not show any sign of overfitting, especially because the early stopping parameter was set to true, even with max_iters being equal to 100. We observe that at lower lr, the model stops early but validation accuracy keeps jumping. At higher lr, the model doesn't converge completely and takes all the 100 epochs. At the ideal lr of 1e-3, train loss decays smoothly while validation accuracy raises and plateaus.